A RELATIVE PROJ OF A FINITE TYPE GRADED ALGEBRA IS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Quasi-compactness of a relative Proj under the same hypothesis. Lemma lemma-relative-proj-quasi-compact (7646-7664, produced as L44246), used at proof lines 7698-7700. Reduces finite typeness to local finite typeness.
- The local characterisation of morphisms locally of finite type. Morphisms, Lemma morphisms-lemma-locally-finite-type-characterize, used at proof lines 7700-7702. Allows the affine-local check.
- Dehomogenisation of a finite type graded algebra. Algebra, Lemma algebra-lemma-dehomogenize-finite-type, used at proof lines 7705-7712. Shows each A_(f) and the twists are of finite type.

Source reference. Original lemma 07Z2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10132–10132. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07ZX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10162–10162. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $\mathcal{A}$ be a quasi-coherent graded $\mathcal{O}_S$-algebra. Let $p : X = \underline{Proj}_S(\mathcal{A}) \to S$ be the relative Proj of $\mathcal{A}$. If $\mathcal{A}$ is of finite type as a sheaf of $\mathcal{O}_S$-algebras, then p is of finite type and $\mathcal{O}_X(d)$ is a finite type $\mathcal{O}_X$-module.*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The assumption implies that p is quasi-compact, see Lemma 07ZX. Hence it suffices to show that p is locally of finite type. Thus the question is local on the base and target, see Morphisms, Lemma 01T2. Say $S = \mathrm{Spec}(R)$ and $\mathcal{A}$ corresponds to the graded R-algebra A. After further localizing on S we may assume that A is a finite type R-algebra. The scheme X is constructed out of glueing the spectra of the rings $A_{(f)}$ for $f \in A_+$ homogeneous. Each of these is of finite type over R by Algebra, Lemma 07Z2 part (1). Thus $\mathrm{Proj}(A)$ is of finite type over R. To see the statement on $\mathcal{O}_X(d)$ use part (2) of Algebra, Lemma 07Z2. $\square$